

Project Closure Report

Project Name: Blockchain voting system **URL:**

<https://projectcatalyst.io/funds/11/cardano-use-cases-concept/blockchain-voting-system> **Project Number:** 1100039 **Project Leads:** David Tacuri **Start Date:** March 2024

Closure Date: March 2026

Project Objective To bring transparency and trust to electoral processes.

List of challenging KPIs and how the project addressed them

- **Use of the Cardano ecosystem in a real case:** A functional electoral system was implemented that registers key evidence on the blockchain, demonstrating transparency, immutability, and auditability in a democratic process.
- **Viability of blockchain in critical processes:** The project validated the use of Cardano in electoral scrutiny, highlighting its security and traceability, as well as limitations in response times.
- **Realistic architectural model:** An architecture was defined in which blockchain functions as a layer of trust and audit, complementing traditional systems without replacing them.

List of project KPIs and how they were addressed

- **System architecture design and implementation:** A complete electoral scrutiny architecture was designed and implemented, including scanning, OCR, digitization, quality control, REST API, and integration with the Cardano blockchain.
- **Development of a functional Proof of Concept (POC):** A fully operational POC was built that executes the complete flow of processing real (anonymized) electoral records, hash generation, data validation, and anchoring of evidence on the blockchain.
- **Integration with smart contracts (Aiken / Plutus):** Plutus validators written in Aiken were developed and tested, incorporating validation rules, access control, and on-chain data flow verification.
- **Backend and REST API implementation:** A robust backend was developed in ASP.NET Core (C#) that interacts directly with cardano-cli to build, sign, and submit transactions, securely coordinating off-chain and on-chain processes.
- **Audit and traceability:** Every transaction contains unique identifiers and structured metadata, enabling public and independent verification of the integrity of the information recorded on the blockchain.

- **Technical documentation:** Complete technical documentation was delivered, including architecture diagrams, smart contract descriptions, API documentation, deployment scripts, and usage examples.

Key Achievements (particularly in technical development and knowledge sharing)

- **Functional implementation of an electoral scrutiny system on Cardano:** The project successfully developed a functional Proof of Concept that demonstrates how the Cardano blockchain can be used as a mechanism for traceability and audit in electoral processes.
- **Generation of reusable technical knowledge:** Relevant technical and architectural lessons were generated that can be reused in other public and private sector use cases where traceability and information integrity are critical.
- **Open-source code and documentation:** All source code, smart contracts, and technical documentation were published in open repositories, contributing to the Cardano open-source ecosystem.

Impact – measurable data and results

- Implementation and validation of a complete end-to-end electoral scrutiny flow.
- Processing and blockchain anchoring of multiple real (anonymized) electoral records.
- Successful compilation, deployment, and execution of smart contracts on Cardano.
- Practical validation of blockchain as a layer of audit and integrity without modifying existing legal processes.
- Identification of real technical limitations and opportunities for improvement within the Cardano ecosystem.

Why is this project important?

This project is important because it provides a realistic and responsible vision of the use of blockchain in critical processes.

Beyond demonstrating that the technology works, the project clearly identified where blockchain adds the greatest value and where it needs to be complemented with other technological layers. In particular, it showed that transaction confirmation times on Cardano's base layer, although secure and reliable, are not optimal for processes that require absolute immediacy, such as an election day under high public pressure.

This type of analysis is fundamental so that the adoption of Cardano in institutional contexts is carried out in an informed, sustainable manner and aligned with the real needs of existing systems.

Next stage of the product / service

Evaluation of Layer 2 solutions (Hydra)

As the next phase of the project, it is planned to study and implement the use of Hydra, Cardano's Layer 2 solution, with the following objectives:

- Significantly reduce response and confirmation times.
- Support high concurrent loads in critical events.
- Maintain the security and integrity of the Cardano ecosystem.
- Enable near real-time operation, with final anchoring on the base layer (Layer 1).

This hybrid approach would allow the system to evolve toward a more scalable model suitable for high-demand scenarios, while maintaining the guarantees of transparency and auditability.

Links to other relevant sources or project documents

Source code:

- Desktop applications: <https://github.com/democracyonchain/asuncion-desktop>
- Transactional backend: <https://github.com/democracyonchain/asuncion-backend>
- Transactional frontend: <https://github.com/democracyonchain/asuncion-frontend>
- Blockchain component: <https://github.com/democracyonchain/asuncion-blockchain>
- Public traceability frontend: <https://github.com/democracyonchain/asuncion-frontend-traceability>
- Traceability backend: <https://github.com/democracyonchain/asuncion-backend-traceability>
- IPFS component: <https://github.com/democracyonchain/asuncion-ipfs>

Documents:

- Technical documents: <https://github.com/democracyonchain/technical-documents>

Closing video:

- Spanish: <https://vimeo.com/manage/videos/1174236453>
- English: <https://vimeo.com/manage/videos/1174237935>